# Week 06: Programming and Research

## Programming task

Session 2: deepen this week’s Python.
Predict → change → repair.
Session 3: apply it to the project.

## Research task

Before Session 1: Research 4 warm-up.
After class: Research 5 on failed updates.
Prepare your evidence; discuss in Week 7.

Open learning_materials/Week_06.ipynb. Follow Start Here; keep your work in that notebook. Email: submit.progmis@gmail.com.

# Programming task

Session 2 · Python practice before project application.

# 1. Predict two objects

## Do this

Open #session-2: Predict → Run.
Read `Counter` before executing.
Predict both `doubled()` results.

## Observe and explain

Identify the class and two instances.
Name `count` and `doubled()`.
Explain which instance is `self`.

# 2. Change one attribute

## Do this

Copy the example.
Set `first.count = 5`.
Predict both returned values; run.

## Observe and explain

Expected: `first` returns 10; `second` 4.
Explain why only one object changes.
Keep prediction and observed output.

# 3. Repair the method

## Do this

Run `BuggyCounter` with `count` 4.
It adds 2 instead of doubling.
Repair `doubled()` in your own copy.

## Observe and explain

Check counts 4 and 1.
Explain why `count` 2 hides the bug.
Optional: compare `alias = first` with `second`.

# Programming practice: done checklist

## Keep in your notebook

Prediction before running
Your changed or repaired code
Observed output or error
One explanation of why it happened

## Next step

Optional exploration can wait.
Save the notebook and a backup.
Continue to the project application.
Submit to submit.progmis@gmail.com.

Outside class: complete the practice and Research preparation.

# Project application: save and recover P11

| Step | What to do / check |
| --- | --- |
| **Open Session 3** | #project-task: use the supplied SQLite setup; record the database path. |
| **Save and reopen** | P11, planning: Q11×2 and Q12×1; save unit snapshots; commit, close, reconnect. |
| **Compare results** | Old estimate 11; change Q11 unit 3→4: current estimate 13, saved estimate still 11. |
| **Check and keep** | Unknown ID handled; stock stays 10/4. Keep rows, relationship sketch, peer note and explanation. |

Save the notebook outputs before cleaning up its temporary database.

# Research database transactions

## What to research

Find out what a transaction is and why related updates should succeed or fail together.

## Possible directions

Use an action with two updates. Consider a failure between them and how to check the records before and after.

## A short summary example

A reservation reduces stock and saves a request. If saving fails, undo the stock change. Compare both records before and after.

Discuss in Week 7 · summarize in your own words and cite your sources.